The Effect of Feedback Loops on Fairness in Rankings from Recommendation Systems

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Abstract. Recently, there has been substantial research undertaken on the role of fairness in recommendation systems. Two causes of algorithmic unfairness are usually identified. Firstly, societal and sampling biases may be inherited from the training datasets into recommendations and secondly, those biases can be further amplified by feedback loops used to update recommendation systems. Although existing literature provides insight on unfairness coming from imbalanced datasets, the long term effects due to feedback loops are still poorly understood. This study seeks to evaluate the impact of feedback loops on author gender fairness in rankings from book recommendations. To answer the research question, rankings from models using several types of collaborative filtering algorithms (*UserKNN, ItemKNN, BPR, and SVD*) and two baselines (*Random, MostPop*) were evaluated in a feedback loop simulation. Further, three notions of fairness in rankings, i.e. *Discounted Cumulative Fairness*, *Fairness of Exposure*, *Equity of Attention* were considered. Obtained findings are inconclusive as the direction and degree of the effect is highly depended on the notion of fairness examined and the choice of collaborative filtering algorithm. Overall, this thesis, draws attention to variables affecting fairness evaluation of a recommendation system such as chosen definition of fairness, learning algorithm in use, ranking positions considered, and the amount of feedback loops already computed.

Keywords: Fairness · Ranking · Feedback loops · Book recommendations.

1 Introduction

Vast advances in machine learning have allowed for the ubiquity of recommendation systems (RSs) in many fields of human endeavor. On a daily basis, platforms such as Netflix, LinkedIn, Trivago or Amazon make use of algorithmic personalization to provide services in order to meet the needs and preferences of their costumers in an efficient manner. Traditionally, evaluation of such models relied mostly on accuracy, diversity, and novelty [41,23]. However, real life examples have brought attention to cases of algorithmic discrimination [45]. For instance, a recent study by [25] showed significant gender imbalance in software engineer positions ad delivery on Facebook that could not be justified by differences in qualifications.

Recent years have witnessed rising academic interest in addressing fairness as a valuable metric of RSs [32,37]. Although the importance of fairness has been recognized, the research to date has not been able to establish a universal definition of fairness. Rather, a multitude of, often mutually incompatible, concepts have been proposed [19]. Most approaches agree, however, that fairness relates to lack of discrimination against sensitive attributes such as gender, religion, age, sexual orientation or race. Discrimination resulting from unfair RSs may not only limit a person's access to varied media content, but also result in financial gains or losses for providers of the ranked content. For instance, being recommended for an Airbnb host means having their apartment rented, and for a book author being recommended translates to having their book sold. In general, lower visibility in a recommendation might limit providers' opportunity to work. This entails a sense of responsibility for RSs creators to ensure that implemented algorithms are objective and free of biases that can lead to discrimination.

It has been widely argued that unfair recommendations might stem from learning algorithms' tendency to sustain societal and sampling biases [44,4] found in training datasets [31]. In an offline quantitative analysis, [17] found that gender imbalances in the music industry translate into track recommendations, as less songs by female artist are recommended in comparison to the ones by male musicians. In real life such behavior of a model can impact financial gains of female artists. Moreover, in some cases, the design of RSs may further amplify biases found in datasets [31]. Personalising systems often use direct feedback loops that learn from user interaction with recommendations and thus, impact the selection of the future training data to improve their accuracy [40]. In the context of recommendations, position bias, according to which users are far more likely to interact with an item on top of the recommendation list [48], is especially important because the system's feedback will likely include top-rated items. In other words, as users interact with recommendations, the system retrains and gives recommendations favoring previously highly positioned items. It is possible, that some biases available in the dataset are then reinforced because of that process [29].

There are two main approaches to evaluate whether or not computed recommendations are fair. Firstly, we can look at the recommendation problem as a classification problem by considering the top-n recommendation list as the positive class. Mehrabi et al. [31] present a comprehensive summary of existing definitions of fairness in machine learning from a classification perspective that could be applied in recommendations. Interestingly, this is the most popular approach observed in the experimental studies [14,50]. Nonetheless, such procedure fails to account for the position bias. Intuitively, an item placed in the top 5 recommendations is at least as valuable as an item recommended in the top 20. This limitation is addressed by following the second approach to evaluate fairness according to which the recommendation problem is considered a ranking problem, where each position is assigned a weight to represent differences in rank. Pitoura et al. [37] introduce a competent taxonomy of fairness notions applicable specifically in rankings, recommenders and rank aggregation.

1.1 Problem statement

Undoubtedly, unfair recommenders are an ethical and societal challenge as they might contribute to discrimination and further division between privileged and underprivileged groups [31]. The aim of this thesis is to gain understanding of how feedback loops in RSs might affect fairness in the long term. More precisely, this work examines author gender fairness in a book recommendation system trained on the BookCrossing dataset. In a related work [14] showed that recommender algorithms tend to translate book author gender distributions found in users' profiles into book recommendation lists. This study evaluates whether or not RSs propagate author gender (un)fairness in book rankings over time due to feedback loops. In conclusion, the study seeks to answer the following research question:

RQ: To what extent do feedback loops impact author gender fairness in rankings computed by a book recommendation system?

For the means of this study collaborative filtering recommendation systems in particular are considered as this type of RSs is prevalent in media recommendations and commonly used in related research [17,46,43,33,14,30]. More precisely, UserKNN, ItemKNN, BPR, and SVD together with two baselines Random and Most Popular are implemented. Further, since the position bias plays a role here, fairness is considered as a ranking problem. To get a comprehensive view of fairness metrics from three different notions of fairness for rankings described in a fairness taxonomy [37] are considered, i.e. *Discounted Cumulative Fairness* [49], *Fairness of Exposure* [42], *Equity of Attention* [5]. *Equity of Attention* refers to individual fairness while the other notions examine group fairness.

1.2 Contribution

The contribution of this thesis is three-fold. Firstly, the effect of feedback loops on author gender fairness is evaluated based on a book recommendation system. In related work, it is a common practise to evaluate model's fairness based on one iteration. However, this approach does not allow for accurate representation of the dynamic behaviour of the system and does not consider the long term effects [12]. Although in this study I take a closer look at author gender the results might start a conversation about other sensitive attributes. Secondly, this work provides a pipeline for evaluating fairness of a book recommender based on ratings from the BookCrossing dataset. The method includes: data preprocessing, author gender detection, recommendation system implementation, feedback loops implementation, and fairness evaluation. In order to facilitate the replication of this study, a Jupyter notebook with full code is accessible here. Thirdly, several notions of ranking fairness, including both group and individual fairness, are discussed in detail and implemented in the experiment. All fairness notions are taken from a single fairness taxonomy by [37] in order to contribute to the effort of the research community to understand and unify existing methods.

It is important to mention that the aim of this study is not to make descriptive nor normative claims about the current state of author gender representation in publicity and book recommenders. The task is rather to provide insight on the behaviour of the algorithms towards possible metrics for measuring fairness.

2 Background

This study sets out to investigate the impact of feedback loops on author gender fairness in rankings from book recommendations. For a better understanding of the problem, sections below discuss the theory behind recommendation systems and fairness concepts used in this study.

2.1 Recommendation Systems

Over the last years, consumers' choices have multiplied exponentially. Intuitively, the aim of a recommendation system is to facilitate users' decision by analysing their preferences and predicting which of the available items are most likely to meet those preferences. There exist a wide variety of recommendation algorithms and the most common way of categorizing them is by considering the type of data they use to derive user preference predictions. For example, Collaborative Filtering (CF) algorithms consider patterns in ratings between similar users and items (Collaborative Filtering, or CF). Another approach is the one used in Content based algorithms (CB), which takes into account certain details about the items that can be recommended. In the context of book recommendation systems that information might include author's name, book genre, or a book summary. Slightly less popular RSs include, amongst others, those based on the preferences of friends or social connections (social filtering), based on user demographics (demographic-based), or based on contextual data such as time or location (context-aware).

From mentioned alternatives, CF algorithms are the most prevalent due to their simplicity and the fact that no additional user or item meta data is necessary to compute them. Since this thesis examines collaborative filtering algorithms, the subsections below highlights key aspects of CF only (see [27] for a comprehensive view on content based recommendation systems or [28,8,26] on recommendation systems in general). More precisely, I present a general definition of a collaborative filtering recommendation system followed by an overview of existing types of CF recommendation algorithms with focus on algorithms implemented in this work's experiment. Further, I describe the role of feedback loops in RSs and briefly discuss the problem of position bias.

Definition Lets consider a set of users U and a set of items I. A recommendation problem may be defined as a function $g : g(u,i) \implies s_i^u$ that for a user $u \in U$

and item $i \in D$ predicts a relevance score s_i^u.The relevance score is predicted for all items in I and a ranking τ^u for u is computed according to the Probability Ranking Principle [39], i.e. ordering all items with decreasing relevance. Usually, an assumption is made that the ranking presented to the user does not include previously rated items. This, however, is highly dependable on the context in which the RS is built. For instance, while users might want to discover new books from a book recommender, a music recommender might repeat certain songs that fit the preference of a user.

Computation of the relevance score is based on a user–item rating matrix which represents past user-item interactions according to some kind of evaluation measure. From this database an algorithm learns about the preferences of user $u \in U$. Ratings are the most popular form of evaluating user-item interactions in RSs [36]. They can be either implicit, that is collected automatically by the system and refer to user activity with the item or explicit, i.e. a user was asked to provide an opinion on a interval or ordinal scale. Overall, explicit ratings are considered more reliable as users distinguish themselves between items they had more or less preference for. On the other hand, implicit feedback requires reduced effort from the user and thus, provides more data. Interestingly, it has been argued that implicit feedback might be more objective, since ratings are not subject to possible social bias to rank some items higher or lower, but rather represent users' genuine interaction with an item [26].

Collaborative Filtering Recommendation Algorithms As previously mentioned, collaborative filtering algorithms solely examine patterns in user-item interactions without any metadata. CF algorithms can be further divided into those using memory-based methods and those following a model-based approach.

Memory Based Memory based collaborative filtering algorithms follow an assumption that similar users like similar items (user-based) and/or that a user will like items similar to the ones they have already seen (item-based). Firstly a user-item matrix is created where rows represent users and columns represent items, respectively. In other words, each row consists of ratings given by a user $u \in U$ to all items in I. A similarity score is then calculated using an appropriate similarity function such as cosine or Pearson coefficient. Closely related users/items may also be called neighbours, which is why memory based CF are often referred to as neighbourhood-based CFs. The memory based CF algorithms used in this study are UserKNN and ItemKNN.

In the case of a user-based algorithm, when trying to predict a score s_i^u for the user u and item i, the similarity measure is first applied to the rows of the ratings matrix, in order to determine neighboring users of the target user u. The ratings are then created by computing the weighted average of ratings given by related users to i. Alternatively, when using an item-based memory based CF the neighbouring items for i are first found by using a similarity measure between the columns of the ratings matrix. From those neighbours, items that a user has already seen are selected and, again, a weighted average is calculated.

This process is repeated for every user and item a score needs to be calculated for.

Overall, memory based CF due to relative simplicity of calculations can provide greater explainability than model based algorithms [36]. Nonetheless, performing calculations for each user-item pair is argued to be relatively inefficient. Further, a criticism of these methods is that they are more susceptible to data sparsity and cold start problem. Data sparsity problem refers to a situation where the number of items is much larger than the number of items rated by users, which is problematic as unrated items cannot be recommended. Similarly, the cold start problem appears when a recommendation needs to be computed for a new user without any previous ratings. Since there is no information available, it is not possible to compute the similarity measure and users are not provided with satisfactory recommendations. Those issues are addressed in the model based algorithms.

Model Based While memory based CF recommendation systems access the user–item rating matrix directly and calculate a score for each user-item pair, model based CF use the matrix to train a model that makes predictions about the scores. Resulting model is used to compute recommendations without having to access the user–item rating matrix repeatedly. This approach is superior to the memory based CF in terms of efficiency and scalability [36]. There are various examples of model based algorithms, such as Bayesian classifiers, neural networks, dimensionality reduction algorithms, or matrix completion techniques [36]. The recommendation systems that were implemented in this study use Bayesian Personalised Ranking (BPR) [38] and Singular Value Decomposition (SVD) algorithms.

Feedback Loops As pointed out in the introduction of this thesis, live recommendation systems are often regularly updated and retrained in order to adapt to the changing needs of the users. Those processes are commonly refereed to as feedback loops [10]. As shown in Figure 1, the model receives feedback from user's interaction with recommendations from the platform and retrains accordingly. The use of feedback loops has been vigorously challenged in recent years by a number of writers.

In one study, [10] argued that feedback loops contribute to homogenization of user behaviour both at population level (behaviour of all users is more alike) and individual level (behaviour of isolated users resembles that of their nearest neighbours). Further, authors argued that homogenization increased with more cycles of the feedback loops. In another analysis, [29] showed that some biases from the datasets can amplify over time when feedback loops are implemented.

In line with presented work, [12] draw attention to the importance of considering the role of feedback loops when evaluating recommendation systems. According to their analysis, single-step analyses do not represent the dynamic behaviour of the system and do not consider the long term effects.

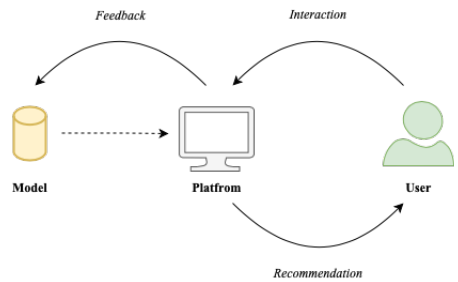

Fig. 1. Visualisation of Feedback Loops. Diagram inspired by the one created in [10]

Position Bias in Recommendation Systems As described in sections above, the output of a RSs for a user is a ranking of items sorted according to calculated relevance. Critics assess the influence of this form of presentation on users and their susceptibility to position bias, according to which users are more likely to interact with items on top of a list [4]. A considerable amount of literature has presented arguments supporting the existence of position bias. In one study participants tended to click on higher positioned items from Google's result page even if mentioned results were less relevant [35]. Similar results were found for users looking at book recommendations from a digital library [11]. Furthermore, in another experiment using eye tracking, participants were less likely to look at lower ranked items in general [21]. It is important to point out that mentioned studies evaluated results on participants from countries in which people read from left to right and from up to bottom. Different results might be obtained when examining users from other cultures with different reading tendencies. Nevertheless, no studies were found evaluating such differences.

The phenomenon of position bias has been translated into mathematical measures and several distributions to calculate a discounting weight for each ranking position are used, such as logarithmic (See Equation 1), geometric (See Equation 2), and singular (See Equation 3) distributions [5,49]. In the geometric distribution p stands for the probability of the user to click on an item. Broadly, the use of discounts is supposed to reflect the decreasing visibility of an item in a ranking. For a better understanding lets consider a ranking τ with 10 items at different positions $j \in [1, 10]$. Figure 2 shows how individual distributions represent visibility at several positions j in the ranking. Furthermore, for the logarithmic and geometric distribution it is possible to perform a cut off at position k. In other words, we can assign 0 values to positions higher than k. This setting is based on an assumption that users will not interact with extremely low-ranked items.

$$w_j = \frac{1}{\log_2(j+1)} \tag{1}$$

$$w_j = p(1-p)^{j-1} \tag{2}$$

$$w_j = \begin{cases} 1, & \text{if } j = 1 \\ 0, & \text{if } j > 1 \end{cases} \tag{3}$$

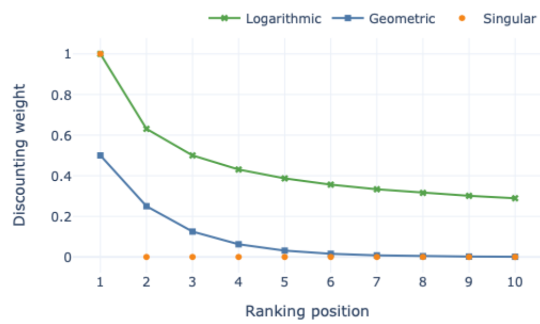

Fig. 2. Discounting weights for ranking positions from 1 to 10.

2.2 Fairness of Recommendation Systems

As mentioned in the introduction to this thesis, recommendation systems might sustain and amplify various types of biases available in the datasets [31]. Broadly, when bias results in discriminating behaviour towards a protected group (later on referred to as G^+) in comparison to a non-protected group (G^-) it is considered unfair. A considerable amount of literature was published with the general aim to translate fairness constraints known in philosophy and politics into algorithmic fairness constraints [51,31,24,6]. Mehrabi et al. [31] present a competent summary of existing fairness constraints in machine learning such as: equalized odds, equalised opportunity or demographic parity (also known as statistical parity). However, just as in philosophy, no universal definition was established and further, existing constraints are often mutually incompatible [19]. For this reason, fairness evaluation of RSs is extremely context depended and each case should be evaluated individually.

The aim of this thesis is not to make any normative claims about fairness, but rather to see how fairness can change over time due to feedback loops. For this reason I focus on different measures of fairness in ranking rather than definitions of fairness constraints. This allows for a more insightful and complex analysis rather than a binary classification of rankings into fair or not fair. In this section, several dimensions of fairness are presented for a brief overview of the fairness problem. Later on, three different measures of fairness in rankings used in this study are described in detail.

Dimensions of fairness When evaluating fairness of a recommendation system there are several perspectives that can be taken. In particular, [37] distinguish between the level and side of fairness in their fairness taxonomy.

Level of Fairness In a discussion on unfair behavior towards minorities it is intuitive to think about a group of people rather than individuals. In a book recommendation setting we might want female authors to, on average, receive equal to male authors visibility in the ranking. Alternatively, we might want to ensure visibility that is proportional to the actual representation of female authors in the population. In both cases, this approach considers fairness from the group level.

Although assessing group fairness is a very common practise in evaluating recommendation fairness [14,17,50], several arguments were made against this approach [7]. The main concern is that by focusing on achieving group fairness, the system might become unfair to a qualified individual by favoring a less-qualified person from the protected group. For this reason the concept of individual fairness was introduced [13]. Fairness on the individual level occurs when similar individuals receive the same outcome, regardless of their belonging to the protected group.

Side of Fairness Common recommendation systems often represent a multi-stakeholder system in which not only the users' needs are considered. [37] distinguish between two classes of stakeholders - consumers and producers. A consumer is the user of the platform, usually an individual that wants to receive some kind of service. Producer on the other hand, is either a person (e.g. job applicant) or creator of an item (e.g. book author), that is being ranked or recommended.

Discrimination may present itself towards stakeholders on both ends of the recommendation system. In particular, [9] identifies three fairness problems relative to the three stakeholder classes: C-fairness related to consumers, P-fairness related to providers, and CP-fairness related to both. In the context of book recommendations, an example of a C-fairness problem could be ensuring that users receive relevant book recommendations regardless of their age. On the other hand, allowing book authors visibility in a recommendation unaffected by their gender refers to the P-fairness problem.

Fairness Measures in Rankings In this section I present several fairness measures for ranking evaluation presented in [37]. As mentioned in the introduction to this thesis, the majority of work on fairness in RSs considers the recommendation problem a classification problem. As shown in previous sections such approach fails to account for the position bias present in user interaction with recommendations. All three metrics presented below highlight this problem and refer to visibility of items in ranking. Two synonymous terms for visibility are used, namely exposure in [42] and attention in [5].

Discounted Cumulative Fairness One approach to account for position in ranking when evaluating fairness is to look at the proportional representation of the protected group at top-p positions of the ranking for various values of p [49]. This method builds on the NDCG@k performance measure (See Equation 10) as it discounts lower positioned items according to the logarithmic distribution, and thus, is referred to as *Discounted Cumulative Fairness* by [37]. In terms of this notion of fairness, a ranking is considered fair if the proportional representation of the protected group at top-p positions of the ranking is identical to the proportional representation of G^+ in the population as a whole.

Normalized discounted difference (rND) in particular is a measure that allows to calculate mentioned proportional representation of G^+ in a ranking τ [49]. More precisely, the rND is computed by accumulating logarithmically discounted difference between the proportion of protected items up to the position p ($|G^+_{1,\ldots p}|$) and their representation in the overall population of size (N). This value is later on normalized using a normalizer Z, i.e. the highest possible value of rND for a given N and $|G^+|$. In general, the closer/further the rND value is to 0/1 the more/less fair the ranking is.

$$rND(\tau) = \frac{1}{Z} \sum_{p=10,20,\ldots}^{N} \frac{1}{\log_2 p} \left| \frac{|G^+_{1,\ldots p}|}{p} - \frac{|G^+|}{N} \right| \tag{4}$$

It is important to mention that this model allows for only binary classification of groups into protected/unprotected. This, disallows fairness evaluation for situations were more than one minority group is discriminated. For example, when we consider ethnicity of a group, several ethnic minorities can be distinguished.

Fairness of Exposure The notion of fairness presented in the previous paragraph, looked at the representation of the protected group at discrete p positions of the ranking. However, as argued by [42], the differences in visibility can be large between all singular positions of the ranking. To address visibility differences at each ranking position j, [42] define $Exposure(G|P)$ (See Equation 6) of a group G in a probabilistic ranking modeled by a doubly stochastic $N \times N$ matrix P. In Equation 6, $P_{,j}$ is the probability that item i is being ranked at position j. The $Exposure(i|P)$ of an individual item i is computed using the Equation 5, where w_j relates to the logarithmic discount at position j. A benefit of this approach is that we can calculate fairness as *Exposure* this for various minority groups, in contrast to *Discounted Cumulative Fairness* where only a binary classification was possible.

$$Exposure(i, \tau) = \sum_{j=1}^{N} P_{i,j} w_j \tag{5}$$

$$Exposure(G|P) = \frac{1}{|G|} \sum_{i \in G} Exposure(i|P) \tag{6}$$

Equity of Attention Similarly to [42,49], [5] address the fact that users interact more with the top-ranked items when assessing fairness. Further, as opposed to other studies mentioned above, their work evaluates fairness on the individual level instead of the group level. In their approach they report on equity of attention, in which a sequence of rankings $\tau^1, ..., \tau^m$ is fair if each item $i \in I$ (where $|I| = N$) receives cumulative attention $\sum_{l=1}^{m} a_i^l$ proportional to its cumulative relevance $\sum_{l=1}^{m} s_i^l$ within that sequence of rankings. They later on compute how much each rating deviates from this fairness assumption to measure *Unfairness of Attention* (See Equation 8).

Attention a_i^l for item i in ranking l is a term synonymous to visibility and is calculated using the geometric distribution w_j to discount position j until a cut-off point k (See Equation 7) for item i at position j in ranking l. Further, relevance relates to the relevance score defined in previous section and can be calculated differently depending on a selected recommender algorithm.

$$w_j = \begin{cases} p(1-p)^{j-1}, & \text{if } j \leq k \\ 0, & \text{if } j > k \end{cases} \tag{7}$$

$$unfairness(\tau^1, ..., \tau^{m)} = \frac{1}{N}\frac{1}{m}\sum_{i=1}^{N}\left|\sum_{l=1}^{m} a_i^l - \sum_{l=1}^{m} s_i^l\right| \tag{8}$$

It is important to mention that this study does not directly focus on measuring fairness in terms of discrimination towards a protected group, but rather looks at general fairness on the individual level. However, this work is still relevant to this thesis as it highlights unfairness given to producers (P-fairness) in rankings.

3 Related Work

This section presents the steadily growing body of literature that attempts to evaluate fairness of recommendations. In subsections below, I first show examples of related experimental works on P-fairness in recommendation systems that either assessed author gender fairness or implemented models trained on the BookCrossing dataset. On the other hand, the second subsection describes a highly relevant study by [30] that evaluated the impact of feedback loops on bias amplification over time.

3.1 Fairness of Recommendation Systems

In a similar work to this one, [14] assessed the impact of collaborative filtering recommender algorithms (i.e. UserKNN, ItemKNN, BPR, and ALS) on author gender distribution of user's book choices. Their fairness evaluation framework is similar to calibration proposed by [43], also mentioned in the taxonomy by [37], as the study compared author gender distribution in users' history with author gender distribution in the top-50 recommendations. Authors discuss that, in general, recommenders adapt to the preferences of users in terms of gender diversity. However, they call our attention to a potential concern that this implies. As a user reads mostly books by authors of one gender they will likely receive recommendations that further amplify that pattern, unless somehow intervened. Nonetheless, the extend of this impact was not explored as no feedback loops were implemented and only one iteration of recommendations was analysed. Work described in this thesis is to some extend inspired by the one in [14] as it also examines author gender fairness in book recommendations. Further, similar methods are used for dataset enrichment as will be discussed in Section 4. However, this study differs as it focuses on fairness metrics in ranking, rather than calibration, and additionally explores the effect of feedback loops.

A different approach to evaluate P-fairness of book recommendations was taken by [20] who examine the effect of production continent on recommendation's disparate visibility and disparate exposure. Disparate visibility, similarly to calibration, measures the amount of times an item is presented in the rankings in relation to the proportion of the item in users' history. On the other hand, disparate exposure takes into account the position bias as it compares exposure of an item in a ranking with its representation in the population. The algorithms in use were UserKNN, ItemKNN, SVD++, BPR, and Biased MF. The study claims that disparities observed in recommendations tend to favor the most represented groups, in this case books produced in Northern America. This study, similarly to this thesis, explores P-fairness using measures considering position bias in book recommendations computed by CF models trained on the BookCrossing dataset. Nonetheless, [20] focus on production continent of a book as a possible sensitive feature, while I examine author gender. Further, different fairness metrics are implemented in this thesis.

In another related study, [17] analysed female artist exposure in music recommendations from Last.fm 360K and LFM-1b datasets. In their experiments, authors examined gender distributions of top-100 recommendations. Moreover, rank of recommendations was to some extend considered by averaging the position of the first occurrence of content by a female for each user. Substantial differences in female exposure where found, as coverage of female artists equalled 28.99%. Further, when analysing top 100 songs recommendations, on average, the first song performed by a female artist was ranked significantly higher than for male artists. Only one algorithm was implemented in this work, namely the ALS algorithm. As shown in [29] different types of algorithms might perform differently, thus, analysis of more algorithms would be an interesting contribution.

As presented several approaches have been taken with regards to related P-fairness in recommendation systems. It is worth mentioning that a vast majority of studies that assessed P-fairness had to carry out an additional data enquiry process in order to gain metadata about the provider [17,14,20]. In the BookCrossing dataset, for instance, while there is information about users' age and nationality, no such information is given about book authors. The dataset is enriched in this thesis as well (more detail in Section 4.3). Further, there is a noticeable discrepancy in used fairness metrics. Although innovation in this field is important, it is also of value to built up on available metrics and unify existing concepts to facilitate their application in the future. This work, examines existing fairness metrics from a fairness taxonomy by [37] to contribute to this unification process. Moreover, individual fairness is assessed, which was not considered in found related work.

3.2 The effect of Feedback Loops on Bias in Recommendation Systems

To date a number of studies reported that RSs might sustain and amplify biases in datasets which can lead to unfairness. Nonetheless, what is still not clear is the long-term effect of this phenomena as a large proportion of studies evaluates only one iteration of recommendations [14,20]. In reality, however, those systems are constantly updated and retrained to provide relevant products. It is key to understand the level of the problem over time in systems that do not use any bias mitigation techniques.

A substantial work has been carried out by [29] who evaluated the impact of feedback loops on bias amplification in a movie recommender based on the movieLens dataset. In their work authors implemented a method for simulating users' interaction with the recommenders in an offline setting and evaluated changes in movie genre distributions in their viewing history over 20 iterations. Overall, authors showed that feedback loops may lead to not only amplified popularity bias, but also declining aggregate diversity and homogenization of the users experience. This in turn might affect P-fairness if the diversity concerns sensitive features, which is relevant from the perspective of this thesis. Further, found effects were found especially strong for users that are a minority in the dataset, which is an important conclusion for understanding the C-fairness problem. In particular, when considering gender of the users, authors observed higher bias amplification for females (which were a minority) than males. Interestingly out of three algorithms considered (UserKNN, BPR, Random), the BPR algorithm showed strongest changes over time. This thesis is to some extent similar to [29] as it evaluates the long term effects of feedback loops. Nonetheless, the focus in this study is on author gender fairness in book recommendations rather than popularity bias in genre in a movie recommender.

4 Dataset Processing and Enrichment

In subsections below all steps taken to transform the dataset are described in detail. Further, Figure 3 represents the general idea behind the tools used within this process.

4.1 Original Dataset

The public BookCrossing dataset used for this study has been downloaded from the website of the University of Freiburg [1]. The original dataset consists of three

[1] http://www2.informatik.uni-freiburg.de/ cziegler/BX/

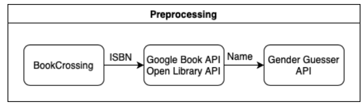

Fig. 3. Dataset Preparation Pipeline.

dataframes about 278,858 users (user dataframe) providing 1,149,780 ratings (rating dataframe) about 271,379 books (book dataframe) identified using ISBN numbers. 62.7% of the ratings are implicit, whilst the rest of the ratings is explicit with values ranging from 1 to 10 (See Figure 4). Moreover, some content-based information about the books that was obtained from Amazon Web Services is provided ('Book-Title', 'Book-Author', 'Year-Of-Publication', 'Publisher'). However, the book dataframe did not initially contain any author metadata necessary to evaluate P-Fairness.

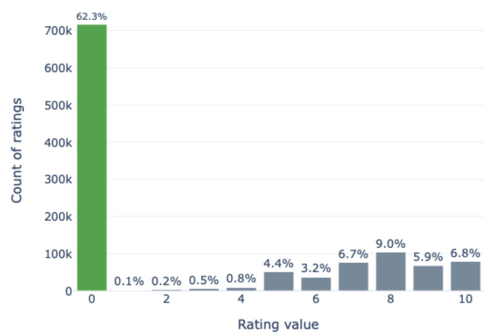

Fig. 4. Distribution of implicit (green) and explicit (grey) ratings in the rating dataset.

4.2 ISBN Linking

Some ISBNs in the book dataframe referred to works with the same author and title, but from different publications. This is unfavorable for two reasons. Firstly, same book can be recommended twice to a user lowering their satisfaction from recommendations. Secondly, such approach might favour larger book publishing companies which have more ratings overall and thus are susceptible to popularity bias [1]. In order to address mentioned issues and decrease the sparsity of the data, related items were linked into single 'books'. This procedure was carried out by querying the Open Library API with an ISBN identifier and collecting information about the *works* attribute which assigns a unique key to linked editions of books. In total, the *works* attribute was found for 92.5% of the book dataset and the linking procedure decreased the number of unique items from 271,378 to 216,496 (decrease of 20,22%). Interestingly, books 'Adventures of Huckleberry Finn' by Mark Twain and 'Frankenstein' by Mary Shelley had over 50 related ISBNs.

A comparable approach of linking ISBNs through the Open Library API was taken by [14]. Similarly to that study, ISBN linking caused some users to have multiple ratings for a book. Nonetheless, this was true only for less than 1% of the ratings. All those duplicates were dropped as will be mentioned in Section 4.4. Another possible limitation of this procedure is that translations of works were linked as well. In other words, 0.7% of books referred to editions in at least two languages. This does not have a direct impact on this study as the aim to is evaluate fairness rather than provide satisfactory service to reader. Nonetheless, it is important to be aware of this implication when implementing ISBN linking.

4.3 Author Gender Detection

The original dataset did not include author gender data, thus, several steps were taken in order to extract this information from the name of the author provided in the dataset.

Author Name Cleaning Due to the fact that the study uses authors' names to predict their genders, data cleaning process was undertaken to facilitate next steps. Firstly, in some cases 'not applicable' was found instead of the author's name. All such examples were changed to $None$ to ease distinction of unnamed authors in the code. Secondly, academic titles such as: 'phd', 'ph d.', etc. were removed from all of the names. Lastly, all names were cleared from punctuation and lower cased.

First Name Detection The first name was detected by splitting the string containing the author's name and extracting the first item. Nonetheless, in some cases (4.8%) the names were abbreviated. In order to gain information about the first name of those authors, Google Book API was used. The API was queried using the ISBN identifier and first author's full name from the *authors* attribute was extracted. The first author was chosen since only the first author is also presented in the BookCrossing dataset. If the first letter of the name and last name from the Google Book API matched the first letter and last name of the author in the book dataframe, the name from the Google Book API was kept. In total, 1.28% of items were updated this way and 3.52% of items remained with abbreviated names.

The Gender-Guesser Package To detect gender from author's name a gender-guesser[2] package from python was used. Although there exist several gender detection packages, this was one was chosen as it supported over 40,000 first

[2] https://pypi.org/project/gender-guesser/

names from not only all European countries, but also from China, India, Japan, U.S.A. The gender-guesser takes a name as input and outputs one of the following labels: 'female', 'male', 'mostly female', 'mostly male', 'andy' (androgynous), or 'unknown' (name not found). Figure 5 shows the representation of detected genders in the book dataframe. Output labels such as 'mostly female', 'mostly male', and 'andy' accounted for a small proportion of all items and were reduced to 'unknown' to improve reliability of the data.

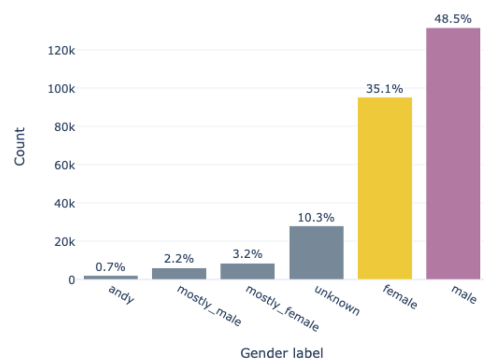

Fig. 5. Gender label representation in the book dataset produced by the gender-guesser.

It is important to mention that this method for author gender metadata acquisition implies certain limitations. Both assuming gender from the name and not accounting authors who are transgender or have a non-binary gender identity is a reductionist approach [22]. Furthermore, it results in the dataset that is not representative towards gender minorities. Moreover, it does not consider authors that use pen names associated with different genders. This is important when considering the fact that in the past women would use male associated pseudonyms to help with their careers [15,18]. It is unknown what percentage of items this problem relates to. Nonetheless, for the purpose of this study the gender detector was chosen to maximise the number of labelled author gender data, since the aim of this work is not to make any claims with regards to author gender, but rather, evaluate the behaviour of recommendation system towards a sensitive attribute. Alternatively, in [14], authors followed VIAF, a reviewed directory of author information compiled from authority records from the Library of Congress and other libraries around the world, as a source to retrieve author's gender. They could identify author gender from around 75% of ratings. In contrast, in this study metadata about over 85% of ratings is collected.

Author Gender Linking As previously mentioned, author gender could not be detected in 3.52% of items due to name abbreviation. Further, in some cases the same author was represented differently in the book dataset, both including and excluding name abbreviation. For instance, the author of the Harry Potter book series was identified as either 'J. K. Rowling' or 'Joanne K. Rowling'. The gender-guesser detector would then assign female gender to only those items that included the name Joanne. To approach this it was assumed that all unique books had the same author. Later, the most common gender label from all related items was chosen to represent overall author gender of a unique book.

Validation To evaluate the accuracy of the gender-guesser detector, 100 unique authors with either male or female gender were randomly selected and manually evaluated by comparing gender assigned by the gender guesser with that from the VIAF website [3]. Authors were searched by name and book title from the dataset. In total 80% of gender labels were correct and 20% where not possible to match as either author was not found in VIAF or no information about author gender was provided.

4.4 Data Selection

Several steps reducing the amount of ratings were taken to increase reliability of the data, but also account for some computation limitations. Table 1 represents a general overview of changes made to the original ratings dataframe with their impact on the overall number of ratings, unique users, and unique ISBNs. Furthermore, information about the number of unique books (after ISBN linking) and gender distribution is given where applicable.

Step 1 Firstly, over 10.3% of ratings contained an ISBN that did not refer to any items in the book dataframe. As a result, gender information about author was lacking as well. For this reason all of the mentioned ratings were dropped.

Step 2 As mentioned in Section 4.2, ISBN linking resulted in a small proportion of duplicated items in the rating dataframe (less than 1%). All duplicates of a rating for the same book were dropped. As will be shown in **Step 4** only implicit ratings (without a score) were used, thus, the scores of the dropped items were not considered.

Step 3 The second step focused on improving the reliability of gender data. Metrics used to evaluate *Discounted Cumulative Fairness* allow only for binary classification of data (protected versus unprotected). Since some of the gender values (14%) were unknown, merging them with either protected or unprotected classes was considered unreliable. To avoid this problem, all ratings related to books with unknown gender were dropped.

[3] https://viaf.org/

Step 4 As mentioned earlier, the original dataset consisted of both implicit (62.7%) and explicit (37.2%) ratings. Further, as shown in Figure 4 the explicit dataset is imbalanced as includes mostly high ratings. For the means of this experiment it was decided to only use ratings with the implicit feedback to allow for better density of data.

Step 5 Finally, to avoid the data sparsity and cold start problems [3] described in Section 2, and consider memory limitations of the study, thresholds were established following the procedure in [34]. More precisely, only users that interacted with over 10 and less than 200 book and only books that were interacted with at least 10 users were considered. This caused a significant decrease, as the majority of users (88.4%) interacted with between 0 to 10 books.

Table 1. Representation of changes in the dataset at each step of the data selection procedure.

	Original	Step 1	Step 2	Step 3	Step 4	Step 5
Ratings	1,149,780	1,031,175	1,023,923	880,105	552,204	57,602
Unique Users	105,283	92,107	92,107	84,681	48,253	2,319
Unique ISBN	340,556	270,170	269,378	226,640	169,960	6,202
Unique Books	-	-	235,844	196,728	148,904	2,306
Female-Author Book Ratings (%)	-	-	407193 (39.7%)	407,193 (46.3%)	263,991 (47.8%)	26,177 (45.4%)
Male-Author Book Ratings (%)	-	-	472,912 (46.2%)	472,912 (53.7%)	288,213 (52.2%)	31,425(54.6%)
Unknown-Gender Book Ratings (%)	-	-	143,818 (14%)	-	-	-

4.5 Effects of Data Selection on Gender Distribution

As shown in the previous subsection several data selection choices were made with regards to the original dataset. Mentioned reductions can have an impact on observed variables, in the case of this thesis - author gender distribution. For instance, Figure 6 displays how female-male author ratings ratio change depending on the evaluated type of ratings (described as **Step 4** in previous subsection). In the BookCrossing dataset, the explicit ratings seem to be slightly more imbalanced than the implicit ratings. Further, it is a relatively common practise when evaluating fairness of RSs to establish certain thresholds on the dataset, constraining the minimum amount of books rated by a user and/or the minimum amount users that have rated a book, to avoid the cold start problem [14,34]. In this work this process was described in **Step 5**. As shown in 7 there exist a general tendency that the more books users interact with the higher their average ratio of books written by a female author. Therefore, the choice of the minimal thresholds can change the way users tendencies are used as training data. This is especially important when making descriptive claims about the current state of author gender distribution in publicity. Although for this and some related work, there exist memory and time limitations which enforce data selection, it is important that researchers are aware of this sampling bias.

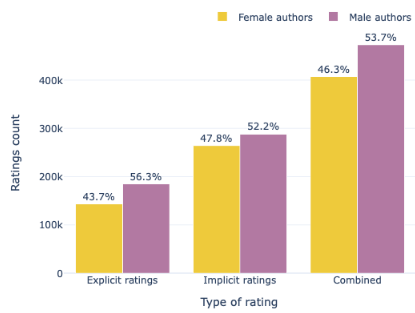

Fig. 6. Ratings count for books written by female and male authors with distinction between explicit, implicit and overall ratings.

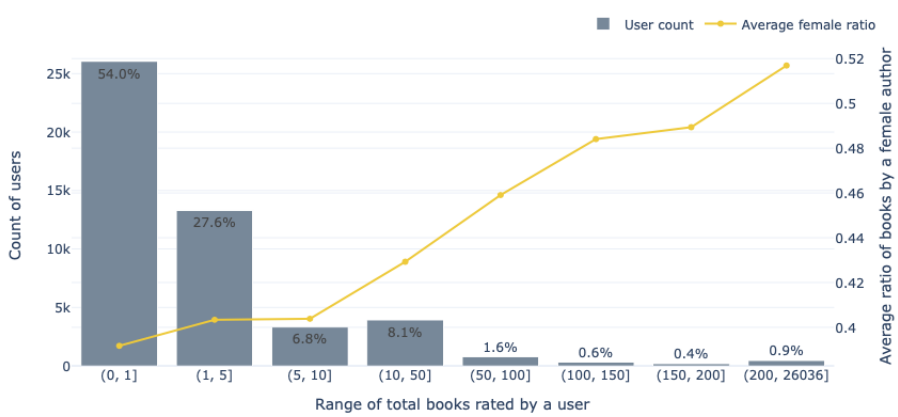

Fig. 7. Average ratio of female authored books per user in implicit ratings depending on the number of books rated by a user.

5 Method

This study takes on a quantitative approach with the aim to understand how do feedback loops in RSs impact author gender fairness. For the purpose of this analysis several models based on collaborative filtering algorithms were implemented, i.e user based KNN (*UserKNN*), item based ItemKNN (*ItemKNN*), Bayesian Personalized Ranking (*BPR*), and a latent factor Singular Value Decomposition (*SVD*). This list consists of algorithms used in related work [30,46,16] but also provides a range of different types of collaborative filtering RSs including both memory and model based algorithms. In addition, similarly to [17,2] algorithms are compared to two baselines, one generating random recommendations (*Random*) and the other recommending most popular books to all of the users (*Most Pop*). All models are initially trained on the BookCrossing dataset (after the pre-processing steps described in Section 4).

An offline simulation is implemented to mimic feedback loops, inspired by the procedure used in [17]. The main experiment is carried out separately for each model as trainings sets change over time and vary depending on the algorithm in use. In other words, different models might create different top10 recommendations that are then added to the dataset. Overall, the experiment consists of the following steps (see Figure 8):

1. The dataset is split into the training and test set using a 80% train 20% test split.
2. The chosen model is trained.
3. Rankings of all items in the training set are computed for each user.
4. Rankings are evaluated with regards to their gender fairness using three notions of fairness, i.e. *Discounted Cumulative Fairness*, *Fairness of Exposure*, and *Equity of Attention*.
5. Top 10 new recommendations are selected from rankings individually for each of the users.
6. Fairness of top-10 recommendations is evaluated using the metric from *fairness of exposure*. Additionally *Coverage* of female authored books is examined.
7. Top-10 recommendations for each user are added to their rating history. New updated dataset is created for the chosen model.
8. The new data set is checked for *Female author ratio*.
9. All steps above are repeated for a total of 10 iterations.

5.1 Cornac Framework

Recommendation algorithms used in this study were implemented using Cornac [4] - a comparative framework for recommender systems available as a python package. This framework consists of a wide variety of recommender algorithms

[4] https://cornac.readthedocs.io/en/latest/index.html

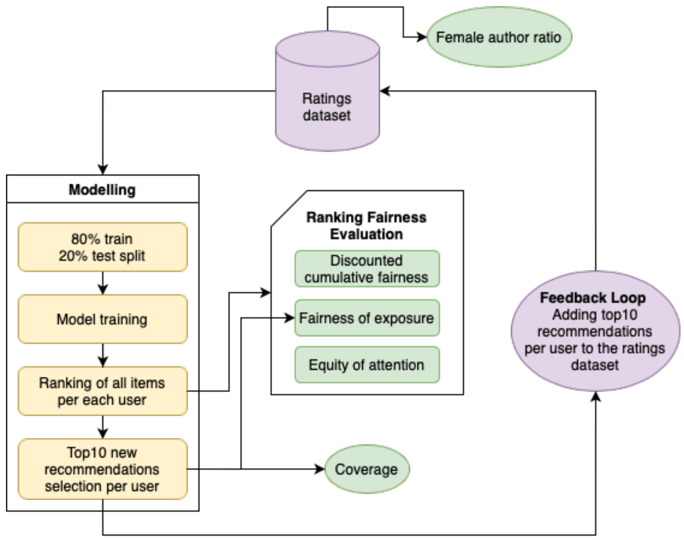

Fig. 8. Experimental pipeline visualisation for one algorithm.

and has been repeatedly used in research related to CF recommendation systems [34,47], thus was considered an appropriate tool for this study.

The dataset had to be transformed in order to meet the format of the Cornac reader. The input consists of a file with only three columns in specific order, i.e. *user_id*, *item_id*, and *rating*. Further, all of the models (besides Random) in the experiment were trained with their default settings from the Cornac package (See Table 2 for details). Lastly, it is worth mentioning that Cornac does not support computing top-n new recommendations. For the purpose of this study, this was additionally implemented by selecting unrated books from the ranking provided by Cornac until a list of top 10 new items was created.

Table 2. Parameters and their values for each algorithm used in the study.

	UserKNN	ItemKNN	BPR	SVD
Number of nearest neighbors	20	20	-	-
Type of similarity	cosine	cosine	-	-
Dimension of the latent factors	-	-	10	10
Maximum number of iterations	-	-	100	100
Learning rate	-	-	0.001	0.01
Lambda regularization	-	-	0.001	0.001

5.2 Metrics

The aim of this thesis is to evaluate the impact of feedback loops on author gender fairness in rankings computed by book recommendation systems. In subsections below I present metrics used to evaluate fairness and traditional performance metrics for additional information.

Fairness Metrics Several notions of fairness are considered in this study, i.e. *Discounted Cumulative Fairness*, *Fairness of Exposure*, and *Equity of Attention*. These notions were chosen following the taxonomy for fairness metrics in rankings by [37] and accounting for the fact that the producer fairness is evaluated. Implementation of metrics to measure each of the notions of fairness is described in subsections below. In all notions of fairness G^+ refers to the protected groups, i.e. female authors, whilst G^- relates to the unprotected group, i.e. male authors. Further, for each iteration and model I additionally measured *Coverage* of female authors in the top-10 recommendations and *Female author ratio* of ratings for female authored books in the overall ratings datasets after each feedback loop.

Discounted Cumulative Fairness In order to assess fairness in terms of *Discounted Cumulative Fairness* the *rND* metric defined in Section 2.2. was implemented following the *rND* function introduced by [49] shown below:

$$rND(\tau) = \frac{1}{Z} \sum_{p=10,20,\ldots}^{N} \frac{1}{\log_2 p} \left| \frac{|G^+_{1,\ldots p}|}{p} - \frac{|G^+|}{N} \right| \qquad \text{(4 revisited)}$$

For the means of this experiment p was set to equal 10, following the experiment in [49]. Further, $|G^+| = 1,074$ and $N = 2,306$. The normalizer Z was calculated using a python code provided by [49] in a github repository [5].

Fairness of Exposure Next, to account for each position in the ranking, rather than each position p, *Exposure* of both the protected (female authors) and unprotected (male authors) groups was measured following the equations introduced by [42]:

$$Exposure(i, \tau) = \sum_{j=1}^{N} P_{i,j} w_j \qquad \text{(5 revisited)}$$

$$Exposure(G|P) = \frac{1}{|G|} \sum_{i \in G} Exposure(i|P) \qquad \text{(6 revisited)}$$

The probability of each item i being placed at position j was set to 1, as deterministic models are considered in this study. Further, the discounting weight w_j was calculated using the logarithmic distribution (See Equation 2) as done by the creators of the measure and $N = 2,306$. The $Exposure(G|P)$ was calculated

[5] https://github.com/DataResponsibly/FairRank

for all rankings for each model in each generation twice. Once with $G = G^+$, and the other time with $G = G^-$. Moreover, to better show differences in exposure in high positions of the rankings, I examine *Exposure@10* of top-10 recommended items. To implement this measure I follow the same equation (See Equation 2) as previously, however, N is set to 10.

Equity of Attention Finally, to assess individual fairness of each model, the *Unfairness of Attention* metric from the *Equity of Attention* [5] notion of fairness was implemented following functions from the original paper.

$$w_j = \begin{cases} p(1-p)^{j-1}, & \text{if } j \leq k \\ 0, & \text{if } j > k \end{cases} \tag{7 revisited}$$

$$unfairness(\tau^1, ..., \tau^{m)} = \frac{1}{N}\frac{1}{m}\sum_{i=1}^{N}\left|\sum_{l=1}^{m} a_i^l - \sum_{l=1}^{m} s_i^l\right| \tag{8 revisited}$$

The probability p of a user clicking on an item was set to 0.5 and the k-cut off was set to 10 to match the size of top recommendations produced. Further, the attention a_i^l of item i in ranking l was calculated according to the weighting function w_j just as in the original study. The relevance score s_i^l was computed using a Cornac ranking function and then normalised. Further, the number of rankings m was set to $2,319$. The *Unfairness of Attention* was calculated for all rankings for each model in each generation thrice. First, to calculate *Unfairness of Attention* for all items regardless of belonging to the protected or unprotected group ($N = 2,306$), second, to compute the metric for G^+ ($N = 1,074$), and third, to calculate the value for G^- ($N = 1,232$).

Performance Metrics In addition to fairness metrics, some of the most commonly used performance metrics were implemented to evaluate the top-10 recommendations such as: Precision@10, Recall@10, Normalised Discounted Cumulative Gain at top 10 positions (NDCG@10) [28]. Those three metrics in particular were mentioned by [10] to better evaluate the real-world performance of RSs. Recall@10 calculates how many books from the test set (relevant books) were recommended to the user in the top-10 recommendations. Further, precision computes how many of the top-10 recommended book were actually relevant to the user. Both recall and precision evaluate recommendations by considering it as a classification problem. On the other hand, NDCG@k measure accounts for the position bias and assesses items relevance proportionate to the ranking position, discounted by the logarithmic distribution, using the following functions:

$$DCG@k(\tau) = \sum_{i=1}^{k} \frac{2^{s_i - 1}}{log_2(i+1)} \tag{9}$$

$$NDCG@k(\tau, \tau*) = \frac{DCG@k(\tau)}{DCG@k(\tau*)} \tag{10}$$

5.3 General Technical Information

All experiments described in this paper were implemented using Python version 3.7.7 and were ran on a MacBook Pro (Retina, 13-inch, Early 2015), with a Dual-Core Intel Core i5 at 2,7 GHz and 8 GB RAM.

6 Results

This section presents all results obtained in this work. Firstly, I show findings that relate to the research question of this thesis, namely, how feedback loops affected each of the three notions of fairness. I return to clearly answer the research question in the discussion in Section 7. Secondly, I describe results from additional analyses that were found relevant for the scope of this paper, namely the differences in computation time for evaluating each of the notions of fairness and the differences in distribution of fairness values for *Discounted Cumulative Fairness* and *Fairness of Exposure*. Lastly, I provide a brief summary of the general performance of all the models implemented in this work.

6.1 Answering the Research Question

In this section I present the overall trends in fairness over time due to feedback loops for all four CF models and two baselines.

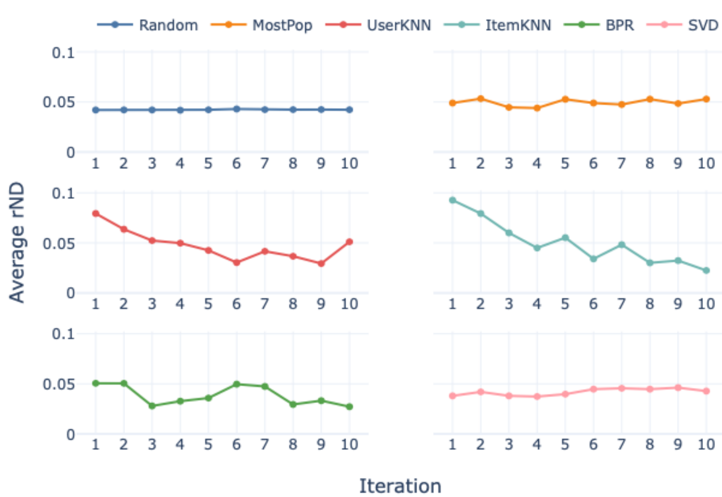

Fig. 9. Average rND value for all six models over 10 iterations. Values closer to 0 represent more fair rankings.

The Effect of Feedback Loops on Discounted Cumulative Fairness Firstly, I examine the effect of feedback loops on fairness measured using the *rND* measure from the Discounted Cumulative Fairness. The measure for all rankings in each of the models was averaged for each of the ten iterations of feedback loops. As shown in Figure 9 values differ depending on the CF model used.

Overall, on average, the most fair rankings in the first iteration were computed by the SVD model, where average $rND = 0.04$. On the other hand, the *ItemKNN* on average produced the least fair rankings with average $rND = 0.09$. In the 10th iteration of the simulation, the most and least fair rankings were given by the ItemKNN (average $rND = 0.02$) and MostPop (average $rND = 0.05$) respectively.

Figure 9 shows gradual decline in the average *rND* value for the memory based models. For both *UserKNN* and *ItemKNN* the average *rND* value was higher than baseline and other models in the first iteration. However, these values decrease to 0.05 and 0.02 in the 10th iteration for *UserKNN* and *ItemKNN* respectively. In other words, rankings computed by those models became slowly more fair with regards to author gender over time. Further, average *rND* value fluctuates around 0.05 for the *BPR model*. No clear dynamic behavior can be found for other models.

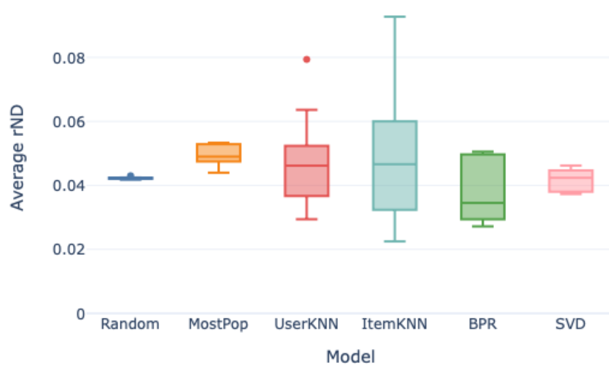

Fig. 10. Box plot of average rND value for all six models from 10 iterations. Values closer to 0 represent more fair rankings.

For a better understanding of general fairness distribution over the 10 iteration, Figure 10 displays a summary in a form of box plots. When looking at Figure 10, it is apparent that the largest fairness variation was observed for the *UserKNN* model. The most stable model, besides baseline, is *SVD*. Interestingly, the *BPR* model shows relatively the most fair rankings when averaging over all 10 iterations. On the other hand, the *MostPop* baseline computes the least fair rankings overall.

The Effect of Feedback Loops on Fairness of Exposure In this subsection I present the differences in exposure for the protected (female authors) and unprotected (male authors) groups over time due to feedback loops. I first discuss the results for the rankings consisting of all items (*Exposure*), and later discuss the results for top-10 recommendations (*Exposure@10*). The measure for all rankings in each of the models was averaged for each of the ten iterations of feedback loops.

Figure 11 displays average exposure given to female and male authors in rankings produced by all six models in all 10 iterations. Overall, the differences are relatively low as the lowest found value for *Exposure* 0.105 was found for the *MostPop* model in 10th iteration and highest value of 0.108 found value for exposure was found for model *ItemKNN* in the 8th iteration.

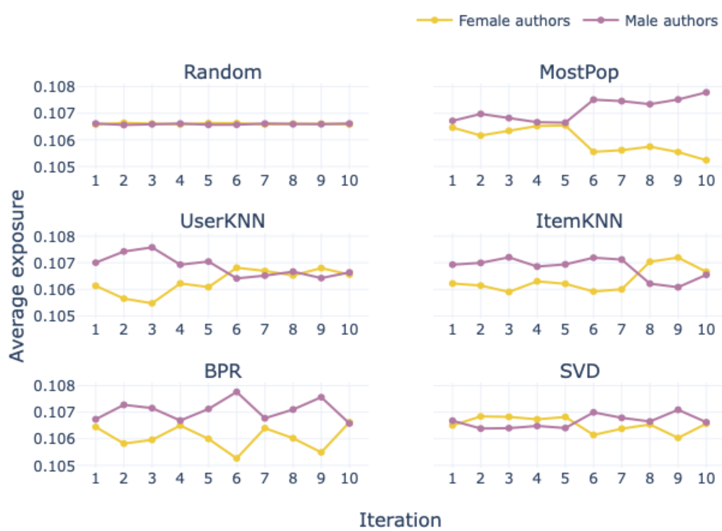

Fig. 11. Average exposure for all six models over 10 iterations with distinction between the exposure for the protected group (female authors) and the unprotected group (male authors).

When considering the trends over time, from Figure 11 we can see that for all models more exposure is given to books by male authors in the first iteration. The difference continues to grow from the 6th iteration for the *MostPop* model. Interestingly, the exposure becomes visibly higher for female authors in comparison to male authors for the *ItemKNN*. Fairness for the *UserKNN* evens out around the 6th iteration. The difference in *Exposure* for the *BPR* model fluctuates over time, although female authors consistently receive less exposure.

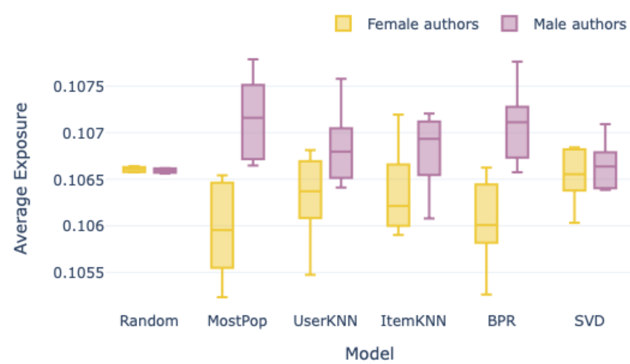

Fig. 12. Box plot of average exposure for all six models from 10 iterations with distinction between the exposure for the protected group (female authors) and the unprotected group (male authors).

From Figure 12 it is apparent that male authors received substantially more exposure than female authors over the 10 iterations for all models but the *SVD* and *Random*. The boxplot further shows that the most extremely low values belong to the protected group, while the highest values belong the unprotected group. Even for *SVD* model this difference is visible from the whiskers of the plot.

Figure 13 compares average exposure given to protected and unprotected groups in the top 10 recommendation. As shown the highest average exposure of 0.630 was noted for female authors in iteration 8 in *ItemKNN* model and the lowest exposure of 0.339 was observed for male authors in iteration 9 in *ItemKNN* model. The average *Exposure@10* in the *MostPop* and *BPR* model is initially higher for female authors than male authors, however, that difference flattens over time. On the other hand, for both of the memory-based models the exposure for both groups is more similar in the first iterations and becomes more divergent with time. Interestingly for both *UserKNN* and *ItemKNN* top recommendations become less fair towards male authors over time. Again, no distinct changes are found for the *SVD* model, in which both groups received comparable exposure over the 10 iterations.

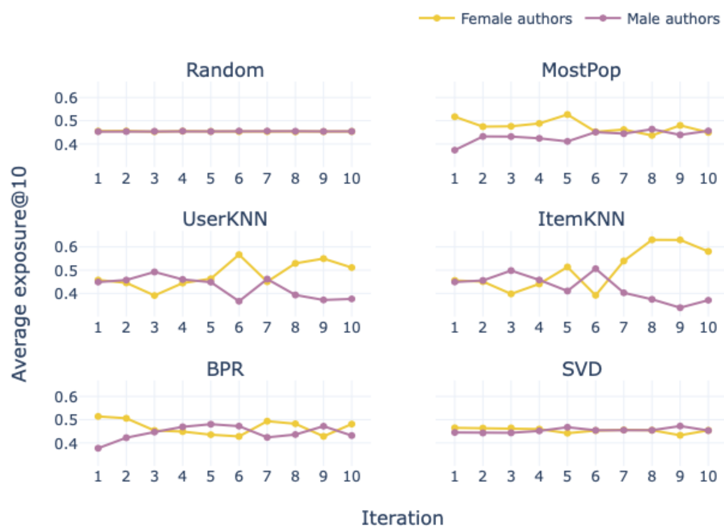

Fig. 13. Average exposure in top-10 recommendations for all six models over 10 iterations with distinction between the exposure for the protected group (female authors) and the unprotected group (male authors).

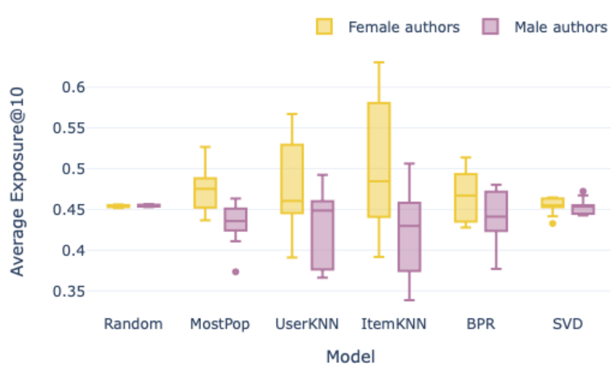

Fig. 14. Box plot of average exposure in top 10 recommendations for all six models from 10 iterations with distinction between the exposure for the protected group (female authors) and the unprotected group (male authors).

Figure 14 provides boxplot of average *Exposure@10* from all 10 iterations. Here, it is more clear that female authors received overall more attention than male authors in the top-10 recommendations. The biggest differences in exposure are found, for the memory-based models and lowest for the *SVD* model and *Random* baseline.

The Effect of Feedback Loops on Equity of Attention In this subsection I describe the effects of feedback loops on individual fairness using the *Unfairness of Attention* metric. Figure 15 shows *Unfairness of Attention* for all items regardless of their author gender. What stands out is that the differences between models are much larger than for other notions of fairness discussed previously. From Figure 15 it can be seen that the most fair rankings were produced by the memory-based models, with *ItemKNN* achieving the highest values for this measure over all 10 iterations. The lowest performance in terms of fairness was observed for the *MostPop* model. No distinct changes over time are observed for models *Random, SVD*. The *UserKNN* decreased in fairness, while *ItemKNN* increased over time. Further, fairness in *BPR* slightly increased over time, while it slightly decreased in *MostPop*.

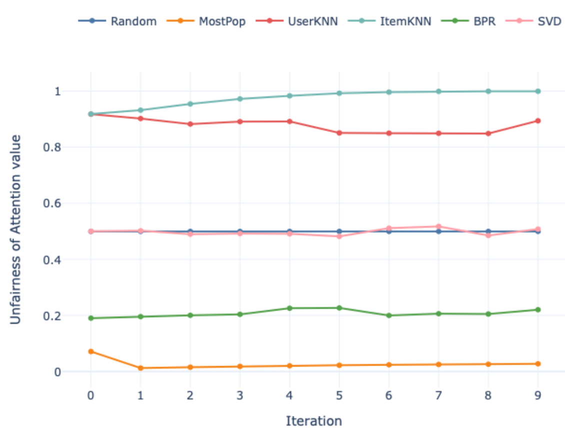

Fig. 15. Unfairness of Attention for all items (protected and unprotected) for all six models over 10 iterations. Values closer to 0 represent unfair rankings.

Plots in Figure 16 display the individual fairness of items depending on the author gender. The smallest difference in *Unfairness of Attention* depending on author gender was observed for *MostPop*. Nonetheless, overall values for this model are significantly lower than for all others. The largest gender differences, on the other hand, was found for *UserKNN* and *ItemKNN*. Lastly, what stands out in presented plots is that books by female authors receive lower scores for all models at all 10 iterations.

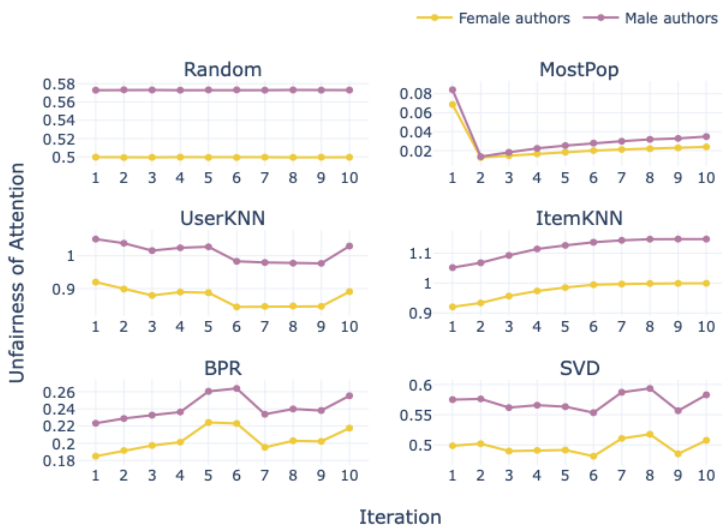

Fig. 16. Unfairness of Attention for all six models over 10 iterations with distinction between the exposure for the protected group (female authors) and the unprotected group (male authors). Values closer to 0 represent unfair rankings. The scale on the y axis is not standardized to better show individual differences.

The Effect of Feedback Loops on Female Author Coverage in Recommendations and Ratio in Ratings In addition to fairness metrics in rankings, I looked at how feedback loops affected the coverage of female authors in the top-10 recommendations and the ratio of female author ratings in the whole dataset.

Figure 17 summarises average coverage of female authored books for all six models in all 10 iterations. The highest average coverage of 6.13 was observed for *UserKNN* in 7th iteration, the lowest of 3.19 for *ItemKNN* in 8th iteration. Average coverage decreases over time for *MostPop*, *ItemKNN* and *BPR* models. For both *MostPop* and *BPR* there is a sharp decrease from the 1st to 2nd iteration. The top recommendations appear to be most balanced for the *SVD* model as average coverage is closest to 5 for most iterations. The boxplot in Figure 18 shows general trends from all 10 iterations. What can be clearly seen is that on average there are more male authored books in top-10 recommendations computed by *Random, MostPop*, and *ItemKNN*. The biggest deviation over time is observed for *UserKNN* and *BPR* models.

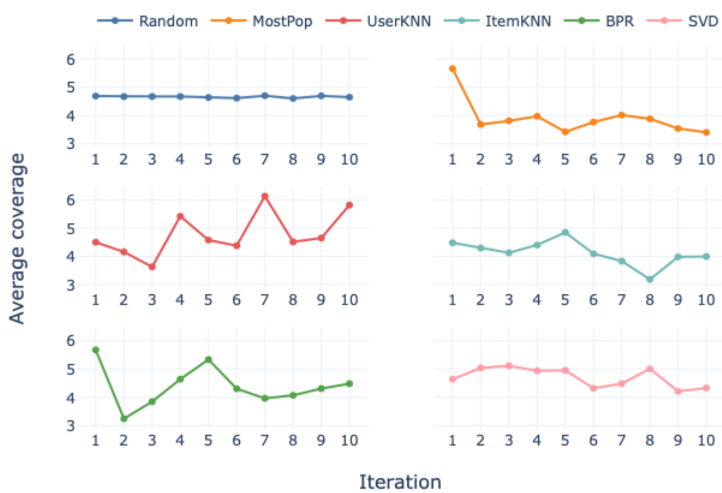

Fig. 17. Average coverage of female authored books in top-10 recommendations for all six models over 10 iterations.

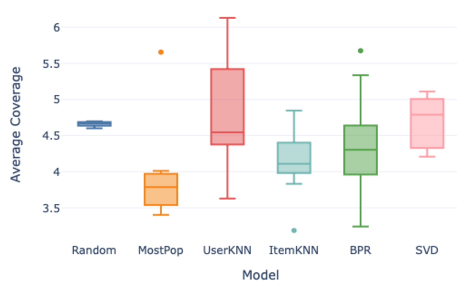

Fig. 18. Box plot of average coverage of female authored books in top-10 recommendations for all six models from 10 iterations.

Figure 19 sets out average ratio of female authored books in the whole dataset. In general, the highest average coverage of 0.49 was observed for *BPR* in the 2nd iteration, the lowest of 0.41 for *MostPop* in the 10th iteration. Further, the graph reveals marked decline over time for *MostPop*, *ItemKNN*, *BPR* models. Interestingly, for both *MostPop* and *BPR* a steep increase is first found in iteration 2. A slight rise in ratio was observed for models *UserKNN* and *SVD*. In general, from box plots in Figure 20 we see that all models except *SVD* result in fewer female writers in the dataset than *Random*.

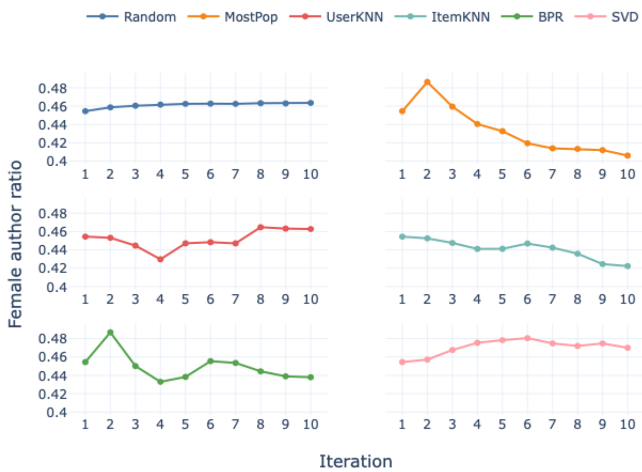

Fig. 19. Ratio of female author ratings in the overall dataset for all six models over 10 iterations.

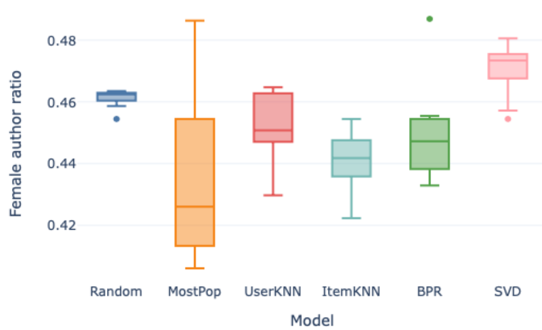

Fig. 20. Box plot of female author ratings ratios in the overall datasets for all six models from 10 iterations.

6.2 Other Findings

For a better understanding of the problem additional analyses were carried out. Firstly, Table 3 represents the time needed to compute fairness evaluation related to each of the three notions of fairness. The overall time to compute all fairness metrics was over 2.5 days. As shown in Table 3 *Fairness of Attention* was the most expensive time wise. Secondly, I looked at the distribution of non-averaged values for metric *rND* and *Exposure*. Figures 21, 22, 23 show decreased deviation of values over time for memory-based models. On the other hand, the *BPR* model shows low standard deviation from the first iteration.

Table 3. Average Computation Time for each Fairness Notion Evaluation per one Model during one Iteration

Notion of Fairness	Average Time per Iteration (sec)
Discounted Cumulative Fairness	878
Fairness of Exposure	100
Fairness of Attention	2682

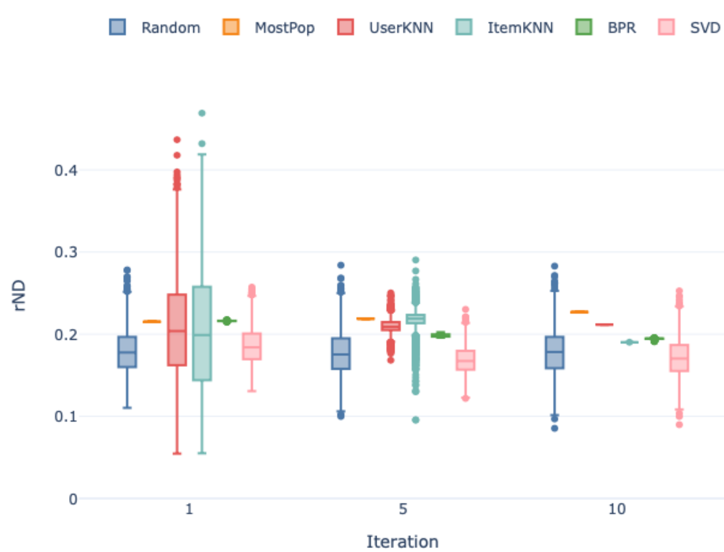

Fig. 21. Box plot of rND values for all six models in all rankings in the 1st, 5th and 10th iteration.

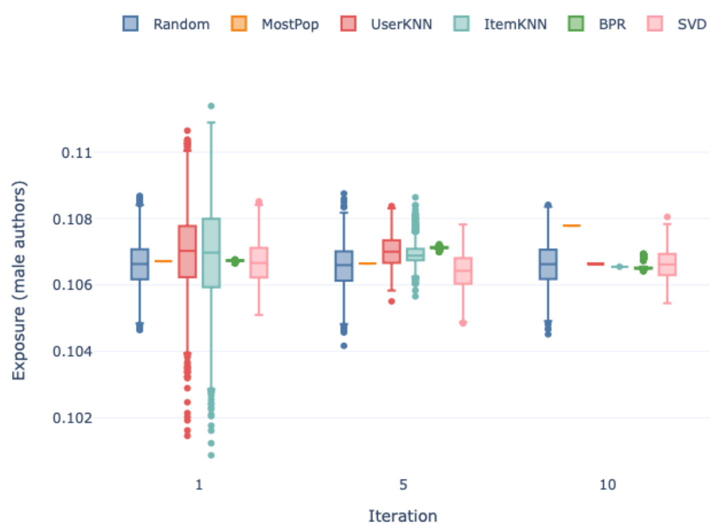

Fig. 22. Box plot of male author books exposure in all rankings for all six models in the 1st, 5th and 10th iteration.

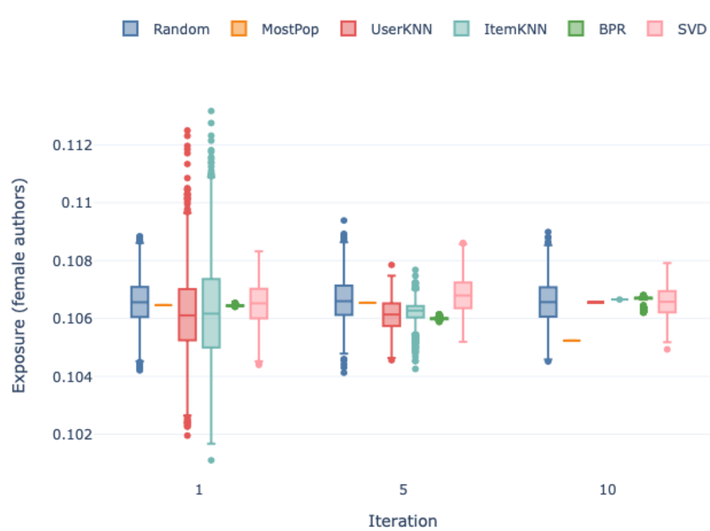

Fig. 23. Box plot of female author books exposure in all rankings for all six models in the 1st, 5th and 10th iteration.

Overall Performance For additional information about the performance of the models used in this study, Figure 24 displays *Recall@10*, *Precision@10*, and *NDCG@10* at each step of the simulation. Most distinct differences were found for *MostPop* and *BPR* models for which plots reveal an increase in the first iteration for all three metrics. Those models also performed best overall. This is followed by a gradual decline for the *Recall@10* and slight decline for the *NDCG@10* metric.

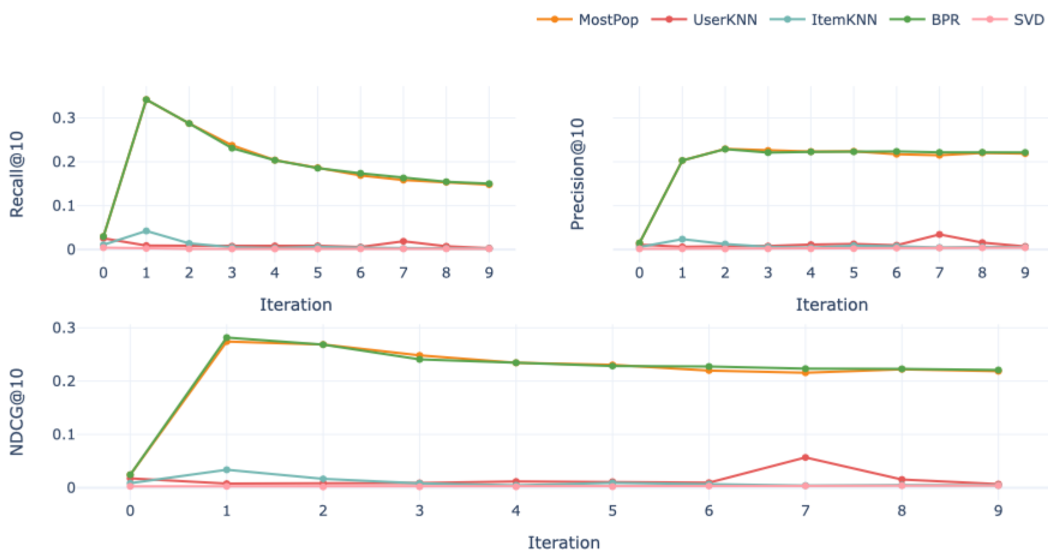

Fig. 24. Model performance using Recall@10, Precision@10, and NDCG@10 over 10 iterations.

7 Discussion

As presented in this thesis, growing body of literature begins to assess fairness of recommendation systems [32,37]. This work in particular discusses the effect of feedback loops on author gender fairness in rankings computed by book recommendation systems. Rankings, from models using several types of collaborative filtering algorithms (*UserKNN, ItemKNN, BPR, and SVD*) and two baselines (*Random, MostPop*), were evaluated using metrics from three notions of fairness in rankings, i.e. *Discounted Cumulative Fairness*, *Fairness of Exposure*, *Equity of Attention*. In this section I first summarize results described in Section 6 to answer the research question and move on to discuss some of the implications of the findings. Secondly, I review limitations relating to this study in Subsection 7.1. Finally, in Subsection 7.2 I provide suggestions for future research.

In general, when evaluating *Discounted Cumulative Fairness* of rankings, memory-based models (*UserKNN, ItemKNN*) became more fair towards female authors over time. However, feedback loops seemed to have lower impact on *MostPop, BPR, SVD* who remained on a similar level of fairness throughout all 10 iterations. On the other hand, the assessment of *Fairness of Exposure* showed long term improvement for *UserKNN*, but deterioration for *MostPop*. Interestingly, ItemKNN initially gives more exposure to male authors but in later stages of the experiment gives more exposure to female authors. The opposite was observed in *SVD*. Different trends where observed when calculating exposure for only top-10 recommendations. In this case, (*UserKNN, ItemKNN*) became less fair with time, while fairness for *MostPop* increased. Finally, feedback loops impacted *Equity of Attention* positively for *ItemKNN, BPR* and *SVD* models, but negatively for *MostPop*.

Broadly, mentioned findings are in accord with [12] who draw attention to the importance of considering the role of feedback loops when evaluating recommendation systems, as changes in fairness were found during the experiments. However, returning to the question posed at the beginning of this study, the extent of the impact of feedback loops on fairness is equivocal as the direction and degree of the effect is highly depended on the notion of fairness examined and the choice of CF algorithm.

The discrepancy in performance of individual models is somewhat in line with the work of [29] who showed different levels of bias amplification due to feedback loops depending on the model. However, in that study the *BPR* algorithm showed strongest changes over time in comparison to *UserKNN* and *Random*. In case of findings presented in this thesis, *BPR* showed moderate changes, while higher impact overall was found for both of the memory-based models regardless of the notion of fairness. Further, fairness of rankings computed by the *MostPop* model was affected when considering *Fairness of Exposure*, but not visibly when assessing *Discounted Cumulative Fairness*. Relatively low impact was found for the *SVD* model for all three notions of fairness. In general, those findings highlight the importance of examining long-term effects on fairness for some of the models. As mentioned previously, the majority of related work [14,20] examines only single-step analyses which do not represent the dynamic behaviour of the system since some models can become more or less fair with time when feedback loops are implemented.

Throughout this thesis the importance of position bias [4] in fairness evaluation is repeatedly underlined. Findings from the simulation show that the effect of feedback loops on *Fairness of Exposure* might be different for the same model depending on the cut-off point of positions examined. In particular, for the *UserKNN* rankings become more fair over time when all items in the ranking are considered. However, this changes when we look only at the top-10 recommendations, as *UserKNN* becomes less fair within the 10th iteration. Further, the differences in exposure for both groups for all models are overall larger when considering only top-10 recommendations in comparison to rankings with all items. This could imply that feedback loops have a stronger impact on fairness of the top recommended items. When evaluating fairness of a recommendation system it is advisable to consider with how many items user is likely to interact with as fairness might be unequal up to different positions in rankings.

Through additional analysis it was found that the standard deviation of non-averaged fairness values for all rankings for *Discounted Cumulative Fairness* and *Fairness of Exposure* decreased over time for *UserKNN, ItemKNN* models, suggesting that with time those models compute rankings of similar level of fairness for all users. The reason for this is unknown, however, it could relate to findings from [29] who found that feedback loops lead to homogenization of the users' experience. Perhaps mentioned homogenization appears also on the fairness level, however, more research is needed to evaluate this phenomenon.

Results from this thesis go beyond the analysis of the effect of feedback loops. One of the most striking findings in this work are the model differences between individual and group fairness. Although all models were computing relatively fair rankings from the group fairness perspective, a large discrepancy was observed in individual fairness, where models were found close to both extremes of the fairness scale. Interestingly, *BPR* was one of the most fair models according to the *Discounted Cumulative Fairness*, but found least fair when looking at *Equity of Attention* (besides the *MostPop* baseline). These findings have two possible implications. Firstly, they highlight the importance of testing several kinds of models in experimental work to conclude about individual fairness in recommendation systems as different models can produce various results. Secondly, they show that fairness in rankings according to one measure does not imply fairness with regards to another. This is in line with [19] who argued that some fairness definitions are mutually incompatible. This discrepancy in fairness depending on the level of fairness becomes problematic when considering arguments by [7], who claimed that focusing on achieving group fairness might be harmful towards individual items that might be positioned lower in spite of high relevancy. When ranked items are in fact people, it is especially important to thoroughly evaluate which level of fairness should be aimed to achieve and choose the best performing model accordingly.

As shown in Section 2.2 the choice of fairness metric might be influenced by the level and side of fairness in focus, but also the (dis)ability to allow for non-binary group classification. As mentioned, evaluation of *Discounted Cumulative Fairness* for more than one minority group is not possible. Further, through additional analysis this work found computation time differences for all three notions of fairness, with *Equity of Attention* found to be most expensive to evaluate.

7.1 Limitations

There are several limitations with regards to this study that need to be discussed. I first draw attention to limitations concerning the dataset enrichment processes described in Section 4. Secondly, I summarize limitations relating to the method and experimental set up (Section 5) and results (Section 6).

Data limitations The method of author gender data enquiry is problematic on three levels. Firstly, assuming gender from the name and not accounting authors who are transgender or have a non-binary gender identity has been criticised as a reductionist approach [22]. It is important to take caution when using this approach, especially when making any claims about the represented group in real life. For this study, no such claims were made and author gender was only used to evaluate the change in algorithmic behaviour over time. In general this limitation points out a gap in the way author data is stored as scholars in related work also found difficulty in choosing a source that supports flexible gender identity records of authors [14]. Secondly, the gender detector was applied on the first name of the book author that was included in the original BookCrossing dataset. However, this information is limited as in case of multiple authors only first author was shown. Again, when making statements regarding the state of author gender fairness in publicity this is highly unfavorable as it does not fully capture existing distribution. However, as mentioned this does not apply to this thesis. Thirdly, as described in Section 4.3, some authors use pen names and thus this method is not reliable in those cases. Notwithstanding these limitations, the use of the gender detection package has contributed to a higher coverage of known gender ratings in comparison to related work [14]. Further, as mentioned in Section 4.3 steps to validate gender labels were taken.

Another limitation of this thesis relates to data selection process described in Section 4.4. Firstly a proportion of ratings with ISBNs that did not have a corresponding author name were dropped. Further, as discussed in Section 4.5 decision between the two types of ratings and establishment of thresholds have impact on gender distribution in the dataset. Perhaps different results could be obtained if alternative choices had been made and/or information about authors' names were found for ISBNs with incomplete information. Nonetheless, whilst the results from this thesis are depended on selection choices, they add to our general understanding of the complexity of the problem.

Experimental limitations Thresholds established in Section 4.4 were chosen partially to avoid the cold start problem in recommendation systems. However, to allow for performance evaluation, the Cornac framework randomly splits the dataset into training and test set by keeping ratings from individual users in both sets. Although unlikely, it is possible, that the cold start problem was not avoided for some of the users. Moreover, caution must be applied when discussing the specific effects of feedback loops on fairness for all six models as only one iteration of the experiment was carried out. Especially when looking at *Fairness of Exposure*, differences between models are relatively small, and perhaps would vary if they were averaged over a larger amount of repetitions. Nonetheless, this thesis contributes to the body of literature by showing that there generally exist differences in fairness, due to feedback loops, between different models. Although the degree and direction could change if different methods had been applied, it is rather clear that some differences might exist.

7.2 Future Research

This thesis is only a step in the research on the effect of feedback loops on fairness in RSs and several opportunities for future research were identified along the way, namely:

- To maximise the use of BookCrossing dataset, future research might explore collecting book metadata for all ISBNs including multiple authors (where applicable).
- A framework for analysing P-fairness when providers are a combination of both protected and unprotected groups is needed (in case of multiple authors).
- As mentioned in Section 3, all related work on P-fairness that was examined had to carry out additional processes to gain author meta data [17,14,20]. It is advisable for researchers building datasets to reflect on to facilitate analyses for future research.
- Investigation and experimentation into effects of data selection on model fairness is strongly recommended.
- Further studies are required to deeply understand model differences in susceptibility to feedback loops. I suggest experimenting with a larger sample of algorithms and carry out statistical test to find significant differences.
- Additional study could examine the effects of feedback loops on homogenization of fairness.
- In general there is a strong need for unification of existing methods to measure fairness and their applicability depending on the context.

8 Conclusion

Overall, this thesis examined the extent to which feedback loops impact author gender fairness in rankings computed by a book recommendation system. To the best of my knowledge it is the first study to examine this problem. To provide a thorough analysis the experiments included a wide range of fairness notions and collaborative filtering algorithms. In conclusion, findings show that no unequivocal answer may be delivered to the research question as the direction and degree of the effect is highly depended on the notion of fairness examined and the choice of CF algorithm. In general, work described in this thesis emphasizes the complexity of fairness. Importance is drawn to thoroughly evaluating the context of research as differences in fairness evaluation might arise depending on the definition of fairness, model in use, ranking positions considered, and finally, the amount of feedback loops already computed.

In addition to results, this thesis provides a comprehensive assessment of concepts regarding recommendation systems (collaborative filtering algorithms, feedback loops, position bias) and fairness (dimensions and metrics of fairness), and related experimental work. Further, described pipeline for data enrichment and experimental set up is publicly shared in a GitHub repository to contribute to the unification of information on fairness in RSs and facilitate the replication of this study.

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